DAA ASSIGNMENT

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Quick Sort by taking three different pivots and Quick select

CODE: -

```
  GNU nano 7.2
#include <stdio.h>

void swap(int *a,int *b){
    int t=*a;
    *a=*b;
    *b=t;
}

int median(int a[],int l,int r){
    int m=(l+r)/2;

    if(a[l]>a[m]) swap(&a[l],&a[m]);
    if(a[l]>a[r]) swap(&a[l],&a[r]);
    if(a[m]>a[r]) swap(&a[m],&a[r]);

    swap(&a[m],&a[r-1]);
    return a[r-1];
}

int partition(int a[],int l,int r){
    int pivot=median(a,l,r);
    int i=l,j=r-1;

    while(1){
        while(a[++i]<pivot);
        while(a[--j]>pivot);
```

```c
        if(i<j) swap(&a[i],&a[j]);
        else break;
    }

    swap(&a[i],&a[r-1]);
    return i;
}

void quicksort(int a[],int l,int r){
    if(l<r){
        int p=partition(a,l,r);
        quicksort(a,l,p-1);
        quicksort(a,p+1,r);
    }
}

int quickselect(int a[],int l,int r,int k){
    if(l==r) return a[l];

    int p=partition(a,l,r);

    if(k==p) return a[p];
    else if(k<p) return quickselect(a,l,p-1,k);
    else return quickselect(a,p+1,r,k);
}
```

```c
int main(){
    int n,k;
    printf("Enter number of elements: ");
    scanf("%d",&n);

    int a[n];

    printf("Enter elements:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&a[i]);

    quicksort(a,0,n-1);

    printf("Sorted array:\n");
    for(int i=0;i<n;i++)
        printf("%d ",a[i]);

    printf("\nEnter k value for kth smallest element: ");
    scanf("%d",&k);

    printf("kth smallest element = %d",quickselect(a,0,n-1,k-1));
}
```

OUTPUT: -

```
Enter number of elements: 8
Enter elements:
5
9
7
6
4
6
2
0
Sorted array:
0 2 4 5 6 6 9 7
Enter k value for kth smallest element: 5
kth smallest element = 6
```